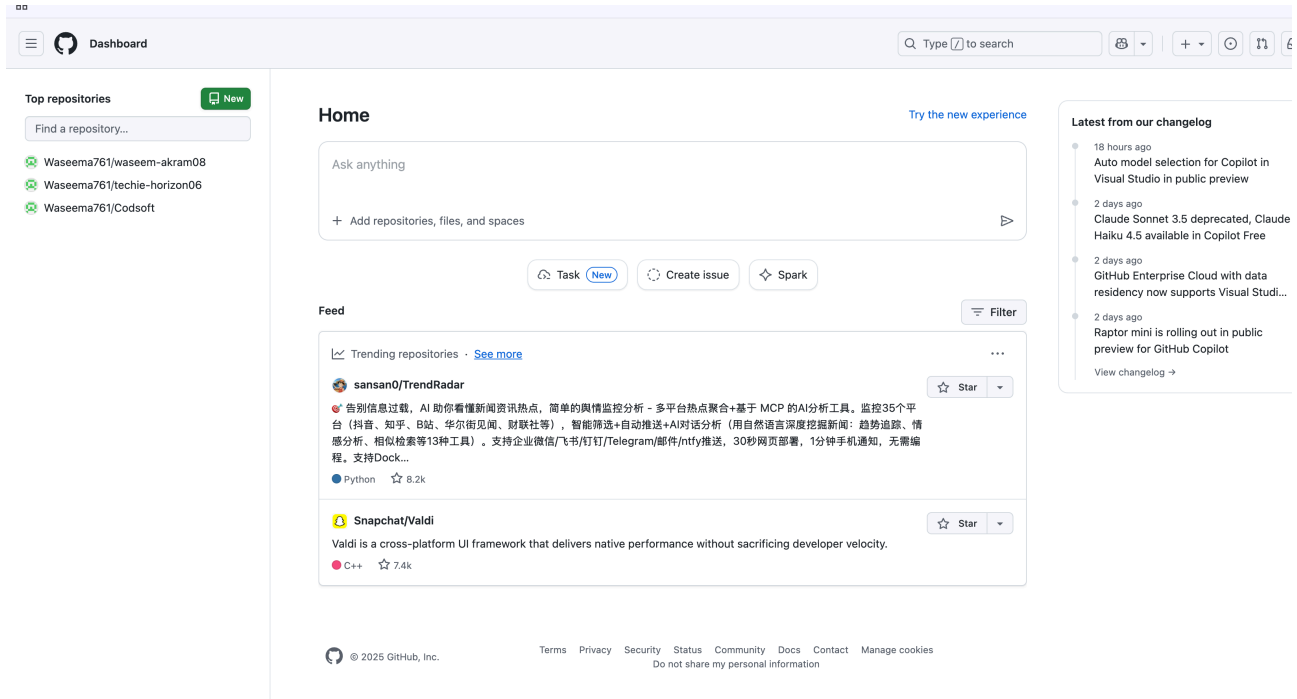


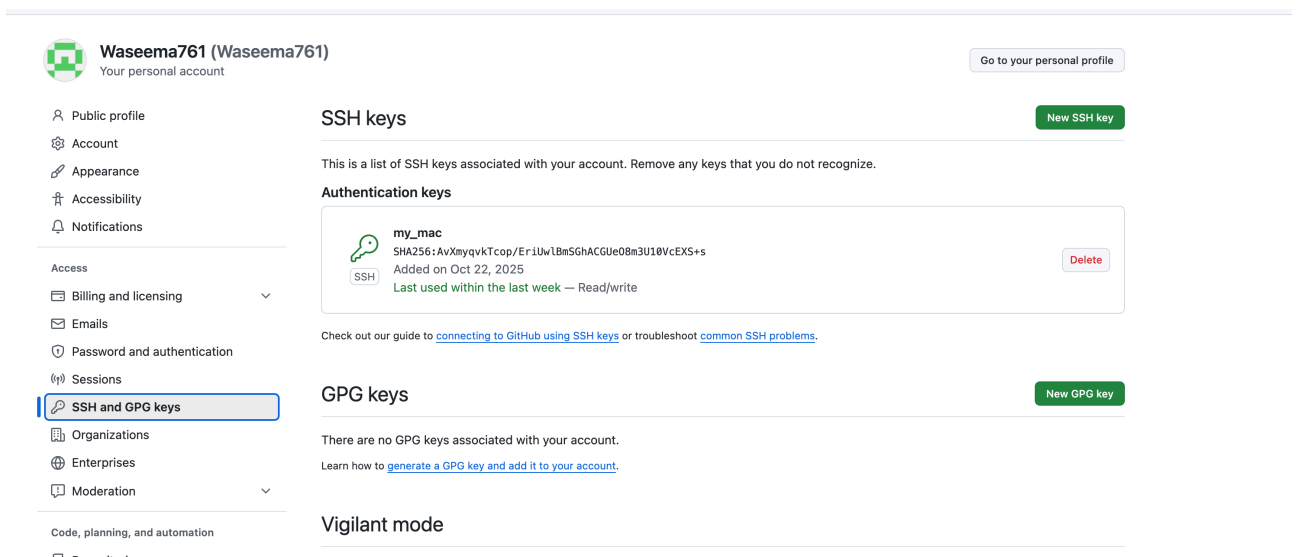
Github task 1

1) Create a GitHub account



Created a account

2) key generated using ssh-keygen for ssh



3) executing all commands from video

Created a directory named Wasim for Git commands execution

```
waseemakram@waseem ~ % mkdir wasim
waseemakram@waseem ~ % cd wasim
waseemakram@waseem wasim % git status
fatal: not a git repository (or any of the parent directories): .git
waseemakram@waseem wasim % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/waseemakram/wasim/.git/
```

git init The command `git init` is used to create a new Git repository in your current directory.

Now inside Wasim named directory we have created a new repository

```
waseemakram@waseem wasim % ls -a
.  ..  .git
waseemakram@waseem wasim % cd .git
waseemakram@waseem .git % ls -alF

total 24
drwxr-xr-x  9 waseemakram  staff  288 12 Nov 15:24 ./
drwxr-xr-x  3 waseemakram  staff   96 12 Nov 15:24 ../
-rw-r--r--  1 waseemakram  staff  137 12 Nov 15:24 config
-rw-r--r--  1 waseemakram  staff   73 12 Nov 15:24 description
-rw-r--r--  1 waseemakram  staff   23 12 Nov 15:24 HEAD
drwxr-xr-x 16 waseemakram  staff  512 12 Nov 15:24 hooks/
drwxr-xr-x  3 waseemakram  staff   96 12 Nov 15:24 info/
drwxr-xr-x  4 waseemakram  staff  128 12 Nov 15:24 objects/
drwxr-xr-x  4 waseemakram  staff  128 12 Nov 15:24 refs/
waseemakram@waseem .git % branch
```

Ls -a to check the repository exist or not because it is hidden

We created a file named Dockerfile

—>Git status :~ The command `git status` is used to check the current state of your Git repository

```
[waseemakram@waseem wasim % git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
[waseemakram@waseem wasim % touch dockerfile
[waseemakram@waseem wasim % git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        dockerfile

nothing added to commit but untracked files present (use "git add" to track)
waseemakram@waseem wasim %
```

We created a file but did not commit it and add that file to repository
Here is how we add that file and commit it

—> `git add Dockerfile` :~ will add the file

```
core:precomposeunicode=true
[waseemakram@waseem wasim % git add dockerfile
[waseemakram@waseem wasim % git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   dockerfile
```

Now have created some multipule files and added all at a time using
—>Git add .

We add all the files at a time

```

[waseemakram@waseem wasim % touch b c d e
[waseemakram@waseem wasim % git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   a
    new file:   dockerfile

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    b
    c
    d
    e

[waseemakram@waseem wasim % git add .
[waseemakram@waseem wasim % git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   a
    new file:   b
    new file:   c
    new file:   d
    new file:   dockerfile
    new file:   e

waseemakram@waseem wasim % █

```

—> `git commit -m "add couple of files"` :~ we have saved the changes
 The command `git commit -m` is used to **save your changes** in Git with a short **commit message** directly in the command line.

```

[waseemakram@waseem wasim % git commit -m "add couple of files"
[master (root-commit) cfb6bc8] add couple of files
6 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 a
create mode 100644 b
create mode 100644 c
create mode 100644 d
create mode 100644 dockerfile
create mode 100644 e
waseemakram@waseem wasim % █

```

List of git commands

...or create a new repository on the command line

```
echo "# techie-horizon06" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin git@github.com:Waseema761/techie-horizon06.git
git push -u origin main
```

...or push an existing repository from the command line

```
git remote add origin git@github.com:Waseema761/techie-horizon06.git
git branch -M main
git push -u origin main
```

—>We have created a repository in our GitHub account and copied the ssh key remote.

The screenshot shows the GitHub interface for a repository named 'techie-horizon06', which is marked as 'Public'. At the top, there are buttons for 'Pin', 'Watch' (0), 'Fork' (0), and 'Star' (0). Below this, there are two main sections: 'Set up GitHub Copilot' and 'Add collaborators to this repository'. The 'Set up GitHub Copilot' section includes a description and a 'Get started with GitHub Copilot' button. The 'Add collaborators to this repository' section includes a search prompt and an 'Invite collaborators' button. Below these sections is a 'Quick setup' section with a header 'Quick setup — if you've done this kind of thing before'. It contains tabs for 'Set up in Desktop', 'HTTPS', and 'SSH'. The 'SSH' tab is selected, showing the repository URL 'git@github.com:Waseema761/techie-horizon06.git'. Below the URL, there is a note about including a README, LICENSE, and .gitignore. The 'Quick setup' section also includes two code blocks: one for '...or create a new repository on the command line' and another for '...or push an existing repository from the command line'. Both code blocks contain the same set of git commands as shown in the previous blocks.

techie-horizon06 Public

Pin Watch 0 Fork 0 Star 0

Set up GitHub Copilot
Use GitHub's AI pair programmer to autocomplete suggestions as you code.
Get started with GitHub Copilot

Add collaborators to this repository
Search for people using their GitHub username or email address.
Invite collaborators

Quick setup — if you've done this kind of thing before

Set up in Desktop or HTTPS SSH git@github.com:Waseema761/techie-horizon06.git

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# techie-horizon06" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin git@github.com:Waseema761/techie-horizon06.git
git push -u origin main
```

...or push an existing repository from the command line

```
git remote add origin git@github.com:Waseema761/techie-horizon06.git
git branch -M main
git push -u origin main
```

—>The command **git remote -v** is used to **show the remote repositories** linked to your local Git project.

—>git remote add origin (ssh url paste here)

The command **git remote add origin** is used to **link your local Git repository** to a **remote repository**

```
waseemakram@waseem wasim % git remote -v
waseemakram@waseem wasim % remote add origin git@github.com:Waseema761/techie-horizon06.git
zsh: command not found: remote
waseemakram@waseem wasim % git remote add origin git@github.com:Waseema761/techie-horizon06.git
waseemakram@waseem wasim % git remote -v
origin  git@github.com:Waseema761/techie-horizon06.git (fetch)
origin  git@github.com:Waseema761/techie-horizon06.git (push)
waseemakram@waseem wasim %
```

Now we have linked our repository

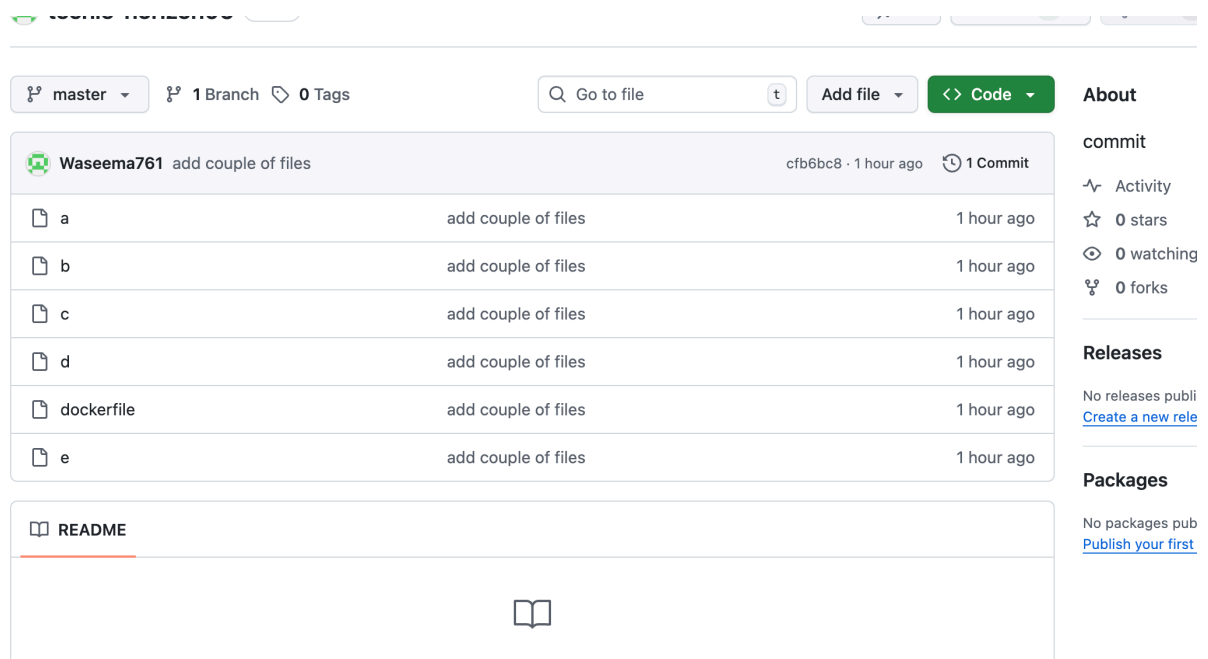
—> git push -u origin master

We have pushed the files from local repository to remote repository

The command **git push origin master** is used to **upload (push)** your local commits to the **remote repository** — specifically to the **master branch**.

```
origin  git@github.com:Waseema761/techie-horizon06.git (push)
waseemakram@waseem wasim % git push -u origin master
[Enter passphrase for key '/Users/waseemakram/.ssh/id_ed25519':
[Enter passphrase for key '/Users/waseemakram/.ssh/id_ed25519':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 239 bytes | 239.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Waseema761/techie-horizon06.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
waseemakram@waseem wasim %
```

Now check the github account repository



We can see all the files from our local repository is available in master branch

—>now we have removed one file

```
branch master set up to track origin/master
[waseemakram@waseem wasim % rm dockerfile
[waseemakram@waseem wasim % ls
a      b      c      d      e
[waseemakram@waseem wasim % git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    dockerfile

no changes added to commit (use "git add" and/or "git commit -a")
waseemakram@waseem wasim %
```

—> In git status we can see the status that one file is deleted

—> git add. And We have to use git commit -m for that changes we have done

Git add .
Git status
Git commit -m

```
waseemakram@waseem wasim % git add .
waseemakram@waseem wasim % git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        deleted:    dockerfile

waseemakram@waseem wasim % git commit -m "deleted dockerfile"
[master 81f412b] deleted dockerfile
 1 file changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 dockerfile
waseemakram@waseem wasim % git push -u origin master
Enter passphrase for key '/Users/waseemakram/.ssh/id_ed25519':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 232 bytes | 232.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Waseema761/techie-horizon06.git
   cfb6bc8..81f412b  master -> master
branch 'master' set up to track 'origin/master'.
waseemakram@waseem wasim %
```

Now we can see that file named dockerfile is deleted from repository

The screenshot shows a GitHub repository page for 'Waseema761/techie-horizon06.git'. The main content area displays a commit titled 'deleted dockerfile' with commit hash '81f412b' and a message 'deleted dockerfile'. Below the commit message, there is a table listing files that were added, deleted, or modified in the commit. The table shows five files (a, b, c, d, e) that were added, each with a message 'add couple of files' and a timestamp of '1 hour ago'. The 'dockerfile' file is listed as deleted. The right sidebar shows the repository's statistics: 0 stars, 0 watching, and 0 forks. The bottom section shows the 'README' file, which is currently empty, with a prompt to 'Add a README'.

File	Change	Time
a	add couple of files	1 hour ago
b	add couple of files	1 hour ago
c	add couple of files	1 hour ago
d	add couple of files	1 hour ago
e	add couple of files	1 hour ago

Now we have created a file named waseem in GitHub repository and now we have to make that available in local repository

 d	add couple of files
 e	add couple of files
 waseem	file on central repo

We used git pull

—> git pull

It will pull that file from remote repository to local

The command **git pull** is used to **download (fetch) and merge** the latest changes from a **remote repository** into your **current local branch**.

```
[waseemakram@waseem wasim % git pull
[Enter passphrase for key '/Users/waseemakram/.ssh/id_ed25519':
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 950 bytes | 237.00 KiB/s, done.
From github.com:Waseema761/techie-horizon06
  81f412b..4da366a master    -> origin/master
Updating 81f412b..4da366a
Fast-forward
 waseem | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 waseem
[waseemakram@waseem wasim % ls
a      b      c      d      e      waseem
waseemakram@waseem wasim %
```

git clone :~The command **git clone** is used to **download (copy)** a **remote Git repository** — like one from GitHub — to your local computer.

```
waseemakram@waseem wasim % git clone git@github.com:Waseema761/techie-horizon06.git
Cloning into 'techie-horizon06'...
[Enter passphrase for key '/Users/waseemakram/.ssh/id_ed25519':
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 8 (delta 2), reused 5 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (8/8), done.
Resolving deltas: 100% (2/2), done.
waseemakram@waseem wasim %
```

We created copy of our repository named techie-horizon06

```
makram@waseem techie-horizon06 % cd ..
makram@waseem wasim % ls
b c d e techie-horizon06
makram@waseem wasim %
```

We have used
git revert :~ for reverting the file which we deleted

```
Revert "deleted dockerfile"
```

```
This reverts commit 81f412b35f296dc431d2d8d370a8d23e1a687e2e.
```

```
# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
```

```
# On branch master
```

```
# Your branch is up to date with 'origin/master'.
```

```
#
```

```
# Changes to be committed:
```

```
#       new file:   dockerfile
```

```
#
```

```
~
```

```
~
```

```
~
```

```
~
```

—>Git log :~The command `git log` is used to **view the commit history** of your Git repository — it shows who made what changes, when, and with what message.

```
[waseemakram@waseem techie-horizon06 % git log
commit 4da366a46d2fc62f1fcddda36bf8476c75f3893a (HEAD -> master, origin/master, origin/HEAD)
Author: Waseema761 <waseem.akram082003@gmail.com>
Date: Wed Nov 12 17:34:15 2025 +0530

    file on central repo

commit 81f412b35f296dc431d2d8d370a8d23e1a687e2e
Author: waseem_akram <waseem.akram082003@gmail.com>
Date: Wed Nov 12 17:25:49 2025 +0530

    deleted dockerfile

commit cfb6bc85f173a4397650e32436255e5a219e4497
Author: waseem_akram <waseem.akram082003@gmail.com>
Date: Wed Nov 12 15:47:29 2025 +0530

    add couple of files
[waseemakram@waseem techie-horizon06 % git revert 81f412
[master 7c7418a] Revert "deleted dockerfile"
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 dockerfile
[waseemakram@waseem techie-horizon06 % ls
a          b          c          d          dockerfile  e          waseem
waseemakram@waseem techie-horizon06 %
```

Git reset :~The command `git reset` is used to **undo changes** in Git — you can use it to move backwards to a previous commit, unstage files, or even erase commits.

```
[waseemakram@waseem techie-horizon06 % touch alphabet
[waseemakram@waseem techie-horizon06 % git add .
[waseemakram@waseem techie-horizon06 % git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   alphabet

[waseemakram@waseem techie-horizon06 % git reset
[waseemakram@waseem techie-horizon06 % git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    alphabet

nothing added to commit but untracked files present (use "git add" to track)
waseemakram@waseem techie-horizon06 %
```